```
pip install gymnasium
```

```
Requirement already satisfied: gymnasium in
/usr/local/lib/python3.12/dist-packages (1.3.0)
Requirement already satisfied: numpy>=1.21.0 in
/usr/local/lib/python3.12/dist-packages (from gymnasium) (2.0.2)
Requirement already satisfied: cloudpickle>=1.2.0 in
/usr/local/lib/python3.12/dist-packages (from gymnasium) (3.1.2)
Requirement already satisfied: typing-extensions>=4.3.0 in
/usr/local/lib/python3.12/dist-packages (from gymnasium) (4.16.0)
Requirement already satisfied: farama-notifications>=0.0.1 in
/usr/local/lib/python3.12/dist-packages (from gymnasium) (0.0.6)
```

```
import gymnasium as gym
```

```
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203:
DeprecationWarning: datetime.datetime.utcnow() is deprecated and
scheduled for removal in a future version. Use timezone-aware objects
to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  return datetime.utcnow().replace(tzinfo=utc)
```

```
import numpy as np
np.bool8 = np.bool_

import gym
import numpy as np

env = gym.make('CartPole-v1')

state = env.reset()

print(f"Action Space: {env.action_space}")
print(f"Observation Space: {env.observation_space}")

def describe_state(state):
    """
    This function prints out the individual components of the state
    State is a tuple (x, x_dot, theta, theta_dot)
    """
    cart_position, cart_velocity, pole_angle, pole_velocity = state
    print(f"Cart Position: {cart_position}")
    print(f"Cart Velocity: {cart_velocity}")
    print(f"Pole Angle: {pole_angle}")
    print(f"Pole Velocity: {pole_velocity}")

print("Initial State:")
describe_state(state)

actions = {0: "Move Left", 1: "Move Right"}
for action in actions:
    print(f"Action {action}: {actions[action]}")
```

```python
num_steps = 5
print("\nSimulating a few steps:")
for step in range(num_steps):
    action = env.action_space.sample()
    next_state, reward, done, info = env.step(action)

    print(f"\nStep {step + 1}:")
    print(f"Action taken: {actions[action]}")
    print("Next State:")
    describe_state(next_state)
    print(f"Reward: {reward}")
    print(f"Done: {done}")

env.close()
```

```
Action Space: Discrete(2)
Observation Space: Box([-4.8000002e+00 -3.4028235e+38 -4.1887903e-01 -
3.4028235e+38], [4.8000002e+00 3.4028235e+38 4.1887903e-01
3.4028235e+38], (4,), float32)
Initial State:
Cart Position: -0.0006804963340982795
Cart Velocity: 0.042463500052690506
Pole Angle: 0.018593482673168182
Pole Velocity: 0.027406521141529083
Action 0: Move Left
Action 1: Move Right

Simulating a few steps:

Step 1:
Action taken: Move Left
Next State:
Cart Position: 0.0001687737094471231
Cart Velocity: -0.15292009711265564
Pole Angle: 0.019141612574458122
Pole Velocity: 0.3258974254131317
Reward: 1.0
Done: False

Step 2:
Action taken: Move Right
Next State:
Cart Position: -0.002889628056436777
Cart Velocity: 0.041924167424440384
Pole Angle: 0.025659561157226562
Pole Velocity: 0.039311863481998444
Reward: 1.0
Done: False
```

```
Step 3:
Action taken: Move Right
Next State:
Cart Position: -0.00205114483833313
Cart Velocity: 0.23666894435882568
Pole Angle: 0.026445798575878143
Pole Velocity: -0.24516606330871582
Reward: 1.0
Done: False

Step 4:
Action taken: Move Left
Next State:
Cart Position: 0.0026822341606020927
Cart Velocity: 0.04117945209145546
Pole Angle: 0.02154247835278511
Pole Velocity: 0.055739808827638626
Reward: 1.0
Done: False

Step 5:
Action taken: Move Left
Next State:
Cart Position: 0.003505823202431202
Cart Velocity: -0.15424466133117676
Pole Angle: 0.022657273337244987
Pole Velocity: 0.35514095425605774
Reward: 1.0
Done: False
```

```python
import gymnasium as gym

envs = gym.envs.registry.keys()

total_envs = len(envs)
print(f"Total number of environments: {total_envs}")
```

```
Total number of environments: 63
```

```python
import gymnasium as gym

envs = gym.envs.registry.keys()

env_names = sorted(list(envs))

for name in env_names:
    print(name)
```

```
Acrobot-v1
Ant-v2
```

```
Ant-v3
Ant-v4
Ant-v5
BipedalWalker-v3
BipedalWalkerHardcore-v3
Blackjack-v1
CarRacing-v3
CartPole-v0
CartPole-v1
CliffWalking-v1
CliffWalkingSlippery-v1
FrozenLake-v1
FrozenLake8x8-v1
GymV21Environment-v0
GymV26Environment-v0
HalfCheetah-v2
HalfCheetah-v3
HalfCheetah-v4
HalfCheetah-v5
Hopper-v2
Hopper-v3
Hopper-v4
Hopper-v5
Humanoid-v2
Humanoid-v3
Humanoid-v4
Humanoid-v5
HumanoidStandup-v2
HumanoidStandup-v4
HumanoidStandup-v5
InvertedDoublePendulum-v2
InvertedDoublePendulum-v4
InvertedDoublePendulum-v5
InvertedPendulum-v2
InvertedPendulum-v4
InvertedPendulum-v5
LunarLander-v3
LunarLanderContinuous-v3
MountainCar-v0
MountainCarContinuous-v0
Pendulum-v1
Pusher-v2
Pusher-v4
Pusher-v5
Reacher-v2
Reacher-v4
Reacher-v5
Swimmer-v2
Swimmer-v3
```

```
Swimmer-v4
Swimmer-v5
Taxi-v4
Walker2d-v2
Walker2d-v3
Walker2d-v4
Walker2d-v5
phys2d/CartPole-v0
phys2d/CartPole-v1
phys2d/Pendulum-v0
tabular/Blackjack-v0
tabular/CliffWalking-v0
```

```python
import gymnasium as gym

env = gym.make("CartPole-v1")

print("Action Space:", env.action_space)
print("Observation Space:", env.observation_space)

state, info = env.reset()
print("Initial State:", state)

action = env.action_space.sample()
next_state, reward, terminated, truncated, info = env.step(action)

print(f"Next State: {next_state}")
print(f"Reward: {reward}")
print(f"Terminated (Goal/Fail): {terminated}")
print(f"Truncated (Time limit): {truncated}")
```

```
Action Space: Discrete(2)
Observation Space: Box([-4.8               -inf -0.41887903        -
inf], [4.8                inf 0.41887903        inf], (4,), float32)
Initial State: [-0.03451778  0.02482397  0.02488642 -0.04741467]
Next State: [-0.0340213  -0.17064583  0.02393812  0.2530151 ]
Reward: 1.0
Terminated (Goal/Fail): False
Truncated (Time limit): False
```